

MISSOURI



MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 1.3

Vectors

Vectors

A matrix with only one column is called a column vector (or simply a vector).

Alternately, a vector is an ordered list of numbers.

Notation for Vectors

In printed materials, vectors are usually denoted by lowercase boldface letters.

Example: $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

In handwritten materials, vectors are usually denoted with an arrow above the lowercase letter.

Example: $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

Real Vectors

The set of all vectors with two entries where both entries are real is called $\mathbb{R}^2$.

The set of all vectors with three entries where all three are real is called $\mathbb{R}^3$.

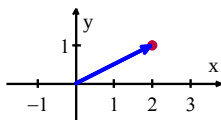
The set of all vectors with n entries where all n are real is called $\mathbb{R}^n$.

Visualizing Vectors

A vector $\begin{bmatrix} a \\ b \end{bmatrix}$ in $\mathbb{R}^2$ can be identified with the point (a, b) in the plane.

It is often useful to think of $\begin{bmatrix} a \\ b \end{bmatrix}$ as the directed line segment from $(0,0)$ to (a, b) .

Example: $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$



Example 1

Let $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$. Find the following quantities both algebraically and graphically.

$$\mathbf{u} + \mathbf{v}$$

$$3\mathbf{v}$$

$$-\mathbf{u} \text{ (equivalent to } -1\mathbf{u}\text{)}$$

$$\mathbf{v} - \mathbf{u}$$

Algebraic Properties of $\mathbb{R}^n$

For all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in $\mathbb{R}^n$ and all scalars c and d , the following eight properties hold.

- 1) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
Vector addition is commutative.
- 2) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
Vector addition is associative.
- 3) $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$
- 4) $\mathbf{u} + (-\mathbf{u}) = (-\mathbf{u}) + \mathbf{u} = \mathbf{0}$

Algebraic Properties of $\mathbb{R}^n$

For all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in $\mathbb{R}^n$ and all scalars c and d , the following eight properties hold.

- 5) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
- 6) $(c + d)(\mathbf{u}) = c\mathbf{u} + d\mathbf{u}$
- 7) $(c)(d\mathbf{u}) = (cd)\mathbf{u}$
- 8) $1\mathbf{u} = \mathbf{u}$

Linear Combinations

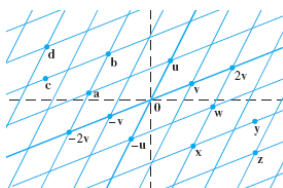
Given vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ in $\mathbb{R}^n$ and scalars $c_1, c_2, \dots, c_p$, the vector $\mathbf{y}$ defined by

$$\mathbf{y} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p$$

is called a linear combination of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ with weights $c_1, c_2, \dots, c_p$.

Example 2

Write vector **c** shown in the figure below as a linear combination of vectors **u** and **v**.



Example 3

Determine if the vector

$$\mathbf{b} = \begin{bmatrix} 11 \\ -5 \\ 9 \end{bmatrix}$$

is a linear combination of the vectors

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix}, \text{ and } \mathbf{a}_3 = \begin{bmatrix} -6 \\ 7 \\ 5 \end{bmatrix}.$$

If it is, find the weights.

Vector Equations

A vector equation

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \dots + x_n \mathbf{a}_n = \mathbf{b}$$

has the same solution set as the linear system whose augmented matrix is

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n \quad \mathbf{b}]$$

Span

If $\mathbf{v}_1, \dots, \mathbf{v}_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_p$ is denoted by $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ and is called the subset of $\mathbb{R}^n$ spanned by $\mathbf{v}_1, \dots, \mathbf{v}_p$.

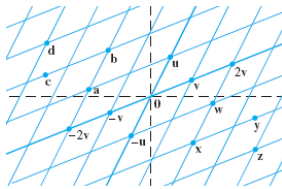
Equivalently, $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is the set of all vectors which can be written in the form

$$c_1 \mathbf{v}_1 + \dots + c_p \mathbf{v}_p$$

where $c_1, \dots, c_p$ are scalars.

Example 2 (revisited)

Give a geometric description of $\text{Span}\{\mathbf{u}\}$ and $\text{Span}\{\mathbf{u}, \mathbf{v}\}$.



Example 3 (revisited)

Determine if the vector

$$\mathbf{b} = \begin{bmatrix} 11 \\ -5 \\ 9 \end{bmatrix}$$

is in the span of the vectors

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix}, \text{ and } \mathbf{a}_3 = \begin{bmatrix} -6 \\ 7 \\ 5 \end{bmatrix}.$$

The Span of One Vector

If $\mathbf{v}$ is a nonzero vector in $\mathbb{R}^2$ (or $\mathbb{R}^3$), then $\text{Span}\{\mathbf{v}\}$ is the set of all points on the line in $\mathbb{R}^2$ (or $\mathbb{R}^3$) passing through $\mathbf{v}$ and the origin.

The Span of Two Vectors in $\mathbb{R}^2$

If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors in $\mathbb{R}^2$, then $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ is either

- a line through the origin and $\mathbf{u}$ if $\mathbf{v}$ is a constant multiple of $\mathbf{u}$.
- All of $\mathbb{R}^2$ if $\mathbf{v}$ is not a constant multiple of $\mathbf{u}$.

The Span of Two Vectors in $\mathbb{R}^3$

If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors in $\mathbb{R}^3$, then $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ is either

- a line through the origin and $\mathbf{u}$ if $\mathbf{v}$ is a constant multiple of $\mathbf{u}$.
- the plane containing the origin, $\mathbf{u}$, and $\mathbf{v}$ if $\mathbf{v}$ is not a constant multiple of $\mathbf{u}$.

Example 4

For what values of h will $\mathbf{y}$ be in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$?

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$$
